Qingyuan Yu

qiy005@ucsd.edu · yuj543226@gmail.com · (619) 997-6003 · La Jolla, CA

Research Interests: Active perception and world models for embodied agents; coupling perception with whole-body control in humanoid manipulation and navigation.

EDUCATION BACKGROUND

University of California, San Diego Sep 2023 – Mar 2027

Bachelor of Science | Expected March 2027

Major: Mathematics-Computer Science

GPA: 3.780 / 4.0 · Provost Honors, 6 quarters

Relevant Coursework: Sensing and Estimation for Robotics (graduate), Deep Reinforcement Learning, Deep Learning, Machine Learning, Autonomous Vehicles, Algorithms, Data Structures, Systems Programming, Modern Algebra, Probability, Mathematical Statistics, Graph Theory

RESEARCH EXPERIENCE

ROAM: Memory-Driven Active Observation for Open-Vocabulary Object Retrieval Jun 2026 – Present

Co-First Author | Advisor: Prof. Xiaolong Wang, UC San Diego

- Designed the perception, memory, exploration, and navigation subsystems; proposed a memory-completeness-driven revisit scheduler that plans multi-degree-of-freedom viewpoints and improved object recall from 28.8% to 46.7%
- Built YOLO-World, SAM2, and 3D association pipelines, and introduced anchor-keyframe-relative pose storage to eliminate accumulated SLAM drift, keeping median object-localization error within 12 cm
- Evaluated the system on GOAT-Bench, improving memory-retrieval success from 0.166 to 0.238 and reducing navigation path length to known targets from 55.1 m to 5.8 m, with a 7.2× improvement in SPL
- Deployed the system on a Dexmate Vega wheeled humanoid robot and designed an adaptive recovery strategy that reduced exploration runs invalidated by getting stuck from approximately 33% to 0%

HuG: Humanoid Whole-Body Affordance-Aware Grasping with Active Perception Nov 2025 – Apr 2026

Co-First Author | Advisor: Prof. Xiaolong Wang, UC San Diego | CoRL 2026

- Proposed the Perceive-Align-Grasp (PAG) embodied-intelligence framework and designed a Bayesian-belief active perception algorithm in the Perceive stage; maintained a probability distribution over a yaw–height viewpoint grid and improved target detection success under severe occlusion from 57.5% to 92.5%
- Bridged perception and downstream whole-body control by aligning active-perception viewpoints online with the input distribution of a CVAE-based grasping policy, and used multidimensional point-cloud quality metrics to determine when to stop observing; stabilized post-detection grasping success at 87.0%
- Integrated Python, PyTorch, ROS2, OpenCV, and point-cloud processing modules; deployed and validated the complete system on a Unitree G1 humanoid robot equipped with a 3-DoF active head and dexterous hands, improving end-to-end success on multi-shelf tasks from 50% to 87.5%
- Trained exclusively on simulation data without real demonstrations and achieved zero-shot sim-to-real transfer, reaching 90% (seen) and 85% (novel) whole-body grasping success rates on the real robot

INTERNSHIP EXPERIENCE

Hikvision Research Institute Jul 2025 – Sep 2025

Machine Learning Intern | Machine Learning Department

- Replaced fixed scoring rules with CLIP cross-modal similarity and reranked image-text results across more than 50,000 product queries, improving image–text ranking accuracy by 4.3% relative
- Processed more than 5 TB of image data and helped build an automated data-quality filtering and checkpoint-based resumable-training pipeline, reducing failed-run recovery time by approximately 60%
- Established structured failure-case tagging and an error-diagnosis workflow, reducing model issue localization from the hour scale to the minute scale and forming a complete diagnosis process across data, algorithms, and experimentation systems

PROJECT EXPERIENCE

Fine-Tuning Qwen3-4B for Reasoning Mar 2026 – Jun 2026

CSE 151B Deep Learning

- Built a three-stage Prompt Engineering, LoRA SFT, and GRPO fine-tuning pipeline, leading the implementation, and improved validation accuracy from 47% to 66.22% under an approximately $50 H100 compute budget
- Addressed training instability caused by inconsistent prompt formats between the SFT and RL stages by designing a prompt-alignment mechanism and hard-prompt boost; quantitatively attributed gains to Prompt Engineering (+12%), SFT (+6%), and GRPO (+1.78%)

Can Vision-Language Models Audit Reward Hacking? Mar 2026 – Jun 2026
CSE 190 Deep Reinforcement Learning

- Designed five reward variants across MuJoCo continuous-control tasks, trained 20 PPO policies, constructed a benchmark, and proposed the hackΔ metric to quantify the severity of reward exploitation
- Analyzed anomalous negative hackΔ values in Hopper and Humanoid, formally distinguishing policy exploitation of reward loopholes from policy non-learning as two failure modes and strengthening the rigor of the VLM audit benchmark
- Strengthened the statistical rigor of RL experiments by analyzing the marginal effects of single random seeds, compute constraints, and high training variance reaching up to a 92% coefficient of variation

Robot State Estimation and SLAM Jan 2026 – Mar 2026
ECE 276A Sensing and Estimation in Robotics (graduate course)

- Derived the linearized point-to-line ICP residual and vectorized k-NN PCA normal estimation with batched eigendecomposition, roughly two orders of magnitude faster than the loop implementation; with GTSAM pose-graph optimization over ~1,000 keyframes, reduced start-to-end trajectory error from approximately 2 m to under 0.1 m on both corridor datasets
- Implemented an EKF on SE(3) with uncertainty parameterized in se(3), deriving landmark and pose Jacobians, and wrote a stereo front-end from scratch (Shi-Tomasi corners, pyramidal Lucas–Kanade tracking, forward-backward and epipolar consistency checks) yielding 6,733 feature tracks over 6,393 frames

Autonomous Driving Car with Vision-Based Traffic Sign Recognition Sep 2025 – Dec 2025
ECE 148 Introduction to Autonomous Vehicles

- Trained an end-to-end CNN behavioral-cloning policy on human-driving demonstrations collected on the vehicle, mapping camera frames directly to steering commands, for trajectory reconstruction
- Built and tuned the team's ROS2 perception-and-control stack on a VESC-driven 1/10-scale vehicle: HSV lane detection, PID steering control, and a four-state machine for blue-tape intersection handling
- Designed and integrated an OCR traffic-sign recognition module that passed real-time textual decision information to the vehicle-control node, achieving 95% recognition accuracy and 10 Hz closed-loop inference in real-world tests

PUBLICATION

X. Peng*, L. Wei*, Y. Zhai*, **Q. Yu***, K. Wang, J. Ye, X. Wang. "HuG: Learning Humanoid Whole-Body Affordance-Aware Grasping with Active Perception." *Conference on Robot Learning (CoRL)*, 2026. Accepted. (*Equal contribution)

EXTRACURRICULAR ACTIVITIES

Chinese Global Community, UC San Diego Sep 2024 – Present
Event Coordinator

- Conceptualized and designed agendas and activity formats for multiple community events, coordinating responsibilities across logistics, promotion, venue setup, and materials; directed on-site execution and established post-event feedback loops that boosted member engagement

TECHNICAL SKILLS

Languages & Frameworks: Python, C++, PyTorch, ROS2, OpenCV, Docker, Hugging Face, GTSAM

Models & Methods: GroundingDINO, SAM/SAM2, YOLO-World, SigLIP, CLIP, LoRA, SFT, GRPO, PPO, Behavioral Cloning, CVAE, EKF and particle filtering, ICP, pose-graph SLAM, point-cloud processing

Platforms & Hardware: Unitree G1 humanoid (3-DoF active head, dexterous hands), Dexmate Vega wheeled humanoid, VESC-driven 1/10-scale vehicle, stereo and RGBD cameras, 2D LiDAR, IMU